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1. Executive Summary

Pakistan's agriculture sector contributes approximately **24% of GDP and employs around half of the country's labour force**, making it essential to economic growth, food security and rural livelihoods. However, climate risks, rising input costs, water scarcity, low productivity and fragmented landholdings continue to constrain the sector, creating demand for accessible and efficient agricultural financing.

Agricultural credit disbursements by the 39 banks covered in this analysis increased from **PKR 1,183.8 billion in 2020 to PKR 2,542.6 billion in 2025**, representing a CAGR of 16.5%. Despite this growth, agriculture's share of total bank advances declined from 7.45% to 7.03%, indicating that the sector has not gained a larger share of overall bank lending.

Commercial banks dominate agricultural financing, accounting for 78.5% of disbursements, while digital banks contribute only 0.05%. This limited digital participation presents a potential opportunity for Digital Retail Banks (DRBs) to serve underserved farmers through cost-effective, technology-enabled lending models.

The **DRB Opportunity Index identifies Punjab as the largest market, scoring 85.5, followed by Khyber Pakhtunkhwa (67.7), Sindh (51.4) and Balochistan (36.2)**. While Punjab offers significant scale, Khyber Pakhtunkhwa presents a particularly attractive entry point due to its large number of small farms and suitability for small-ticket digital financing.

Government initiatives, including the State Bank of Pakistan's Risk Coverage Scheme for Small Farmers and Underserved Areas and Zarkhez-e, provide opportunities to expand agricultural lending through risk-sharing arrangements and digital loan delivery. These initiatives can support improved farmer verification, input-based financing and access to formal credit.

For a DRB, the strategic opportunity lies in developing an integrated digital agricultural finance model offering **crop loans, agricultural input financing, livestock finance, post-harvest finance and value-chain lending**. By combining digital onboarding, alternative-data-based credit assessment, remote verification and partnerships with agricultural merchants, a DRB can reduce servicing costs and improve access to finance. A phased strategy prioritizing small farmers in Khyber Pakhtunkhwa, followed by expansion into Punjab and selected opportunities in Sindh, could support sustainable growth while strengthening financial inclusion.

2. Pakistan’s Agriculture Sector Overview

Agriculture remains a cornerstone of Pakistan’s economy, contributing approximately **24% to GDP** and providing employment to around **half of the country’s labour force**. The sector is critical for food security, rural livelihoods, foreign exchange earnings and a wide range of downstream industries, including food processing, textiles and trade.

2.1. Economic Importance

Agriculture is the largest sector of Pakistan's economy and has strong linkages with both rural and urban economic activity. Its performance directly affects household incomes, food prices, industrial production and exports. The sector also supports millions of farmers and workers across crop production, livestock, fisheries and related value chains.

2.2. Key Crops and Production

Pakistan's major crops include **wheat, rice, cotton, sugarcane and maize**. These crops account for a significant share of cultivated land and agricultural output. However, the growth of major crops has remained relatively weak in recent years, highlighting the need for improved productivity, better inputs, mechanization and greater diversification.

Other high-value crops, including **fruits, vegetables and pulses**, are becoming increasingly important. Mangoes, particularly from Punjab and Sindh, represent an important export-oriented agricultural product.

2.3. Livestock and Other Sub-Sectors

Livestock is a major component of Pakistan's agriculture sector, with dairy, cattle and poultry contributing significantly to agricultural income and rural employment. The livestock sector has also demonstrated greater resilience than some crop segments. In FY2025, livestock recorded **4.72% growth**, supporting overall agricultural sector performance.

Fisheries and forestry also contribute to agricultural output and provide additional sources of income and employment, particularly in rural and coastal communities.

2.4. Key Challenges in Agricultural Borrowing in Pakistan

Pakistan's agricultural lending sector faces several structural challenges that limit farmers' access to formal financing and increase lending risks for financial institutions.

1. **Inadequate Insurance Coverage**
   Limited agricultural insurance exposes farmers and lenders to significant risks from crop failure, extreme weather events, and other agricultural losses. This uncertainty can discourage banks from extending credit to farmers.
2. **Limited Availability of Accredited Warehouses**
   The shortage of accredited warehouses restricts the use of stored agricultural commodities as collateral. This makes it difficult for banks to secure loans against produce and increases the risks associated with agricultural commodity financing.
3. **Limited Access to Formal Financing**
   Many farmers face barriers to obtaining formal credit due to high lending risks, insufficient collateral, and inadequate insurance coverage. These constraints particularly affect small-scale farmers, limiting their ability to invest in modern farming techniques, quality inputs, and productivity-enhancing technologies.

Impact on the Agricultural Sector

These challenges restrict agricultural investment, constrain productivity growth, and limit farmers' income potential. Strengthening agricultural insurance, expanding accredited warehousing infrastructure, and improving access to formal credit are essential to promoting sustainable agricultural development and supporting Pakistan's food security.

3. Modernization and Technology in Agricultural Lending in Pakistan

Agricultural lending in Pakistan has traditionally depended on branch visits, manual land-record checks and collateral, making small-ticket loans costly and hard to scale. Technology is now changing this model.

1. **Digital onboarding** through biometric verification, digital KYC, mobile banking and branchless agent networks allows farmers to access finance without visiting a branch.
2. **Remote verification** is now possible as SBP permits satellite imagery and geo-fencing in place of traditional Khasra Girdawari, while digitized land records such as LIMS speed up ownership and tenancy checks.
3. **Alternative data**, including mobile wallet activity, input purchases and past repayment behaviour, helps banks assess farmers who lack formal credit histories.
4. **Merchant-led disbursement**, as under Zarkhez-e, channels financing directly to verified input suppliers, ensuring productive use of funds and creating a digital transaction record.
5. **Risk tools** such as weather-index insurance, satellite crop monitoring and electronic warehouse receipts help reduce losses and strengthen collateral.

Together, these developments are shifting agricultural lending toward a data-driven, digitally delivered model. For Digital Retail Banks, technology is the key advantage that makes serving large numbers of small farmers commercially viable.

4. Policy and Strategic Outlook

Pakistan's agriculture strategy increasingly emphasizes **productivity improvement, diversification, technology adoption, export growth and private-sector participation**. The development of high-value crops, livestock, modern irrigation, agricultural processing and value-chain infrastructure can help improve farmer incomes and strengthen Pakistan's agricultural competitiveness.

For the banking sector, these structural developments create opportunities across **crop finance, livestock finance, agricultural inputs, farm machinery, irrigation, post-harvest finance and value-chain lending**. Government and State Bank of Pakistan initiatives are also increasingly supporting digital and risk-sharing mechanisms to expand formal agricultural financing.

5. Major Agriculture Financing Schemes

1. **Zarkhez-e (Assan Digital Zarai Qarza)**: A digital collateral-free financing initiative allowing farmers to apply via mobile app for up to PKR 1,000,000 (landowners) or PKR 500,000 (tenants) to purchase seeds, fertilizers, and diesel.
2. **Prime Minister's Youth Business & Agriculture Loan Scheme (PMYB & ALS)**: Provides subsidized loans categorized into Tier 1 (up to PKR 0.5M, interest-free), Tier 2 (PKR 0.5M to 1.5M at 5% markup), and Tier 3 (PKR 1.5M to 7.5M at 7% markup).
3. **Kissan Khushhal Scheme (KKS)**: Offered by specialized institutions like ZTBL to meet medium and long-term development needs of farmers, including land leveling and tube well installation.
4. **Revolving Credit Schemes (HariBhari / Kissan Revolving Credit)**: Provided by banks like Allied Bank and Bank AL Habib as a flexible working capital line for multi-season crop inputs (seeds, fertilizers, pesticides) with one-time documentation.
5. **Electronic Warehouse Receipt (EWR) Financing**: Enables farmers and agricultural operators to secure financing against stowed crops (like rice and maize) kept in accredited professional warehouses.
6. **Tractor and Farm Mechanization Financing**: Conventional and Shariah-compliant (such as Allied Aitebar Ijarah or Sindh Abadgar Tractor Scheme) facilities designed to help farmers purchase tractors, balers, and modern equipments.

6. Risk Coverage Scheme for Small Farmers and Underserved Areas

The **State Bank of Pakistan (SBP), in collaboration with the Government of Pakistan, introduced the Risk Coverage Scheme for Small Farmers and Underserved Areas** to encourage financial institutions to expand agricultural lending to underserved segments. The scheme is effective from **July 1, 2025, to June 30, 2028**, and provides participating lenders with **10% first-loss coverage on the principal amount of eligible agricultural loans**.

The scheme covers financing for **crop production, dairy, livestock, and fisheries**, with financing of up to **PKR 3 million per borrower**. Loans are generally available for up to **12 months**, while financing for sugarcane may extend to **18 months**.

Key Features

Feature	Details
Scheme period	July 1, 2025 – June 30, 2028
Eligible sectors	Crops, dairy, livestock & fisheries
Maximum financing	Up to PKR 3 million per borrower
Loan tenor	Up to 12 months; up to 18 months for sugarcane
Government support	10% first-loss coverage on principal
Participating institutions	Commercial, Islamic, specialized & microfinance banks

The scheme has a targeted geographic approach. In **Punjab and Sindh**, financing is focused on **subsistence and small farmers**, while in **Khyber Pakhtunkhwa, Balochistan, Azad Jammu & Kashmir, and Gilgit-Baltistan**, farmers across different farm and landholding sizes can benefit. This regional structure is intended to encourage lending in areas where agricultural finance remains relatively underdeveloped.

6.1. Zarkhez-e: Digital Delivery Channel

A major digital component supporting the expansion of small-farmer financing is **Zarkhez-e (Asaan Digital Zarai Qarza)**, an SBP-supported centralized digital platform designed to facilitate agricultural loan applications without requiring farmers to visit a bank branch.

Under the Zarkhez-e model, financing is primarily provided **in kind**, with at least **75% of the approved financing** used for agricultural inputs such as seeds, fertilizer and diesel through verified agri-merchants, while up to **25% may be provided in cash**. Farmer and agricultural activity verification can be supported through the **Land Information Management System (LIMS)**.

6.2. Banking Implication

The scheme reduces a key barrier to agricultural lending by **sharing a portion of lenders' credit risk with the government**. Combined with Zarkhez-e's digital application, verification and merchant-based disbursement model, it provides a framework for expanding **small-ticket, digitally enabled agricultural finance**, particularly among farmers who have historically had limited access to formal banking channels.

6.3. Digital Transformation of Agriculture Lending

Pakistan's agricultural lending ecosystem is increasingly moving toward **digital verification, digital onboarding and technology-enabled credit assessment**. The State Bank of Pakistan (SBP) has introduced measures that allow banks to use **digital survey reports based on geo-fencing and satellite imagery** as an alternative to traditional Khasra Girdawari for verifying agricultural activity. This can help reduce manual verification requirements and improve the efficiency of agricultural loan processing.

The development of **Zarkhez-e (Asaan Digital Zarai Qarza)** further supports this transition by providing a centralized digital platform through which eligible farmers can apply for agricultural financing. The platform incorporates digital verification and facilitates input-based financing through verified agricultural merchants.

6.4. Digital Agriculture Lending Journey

Digital Application → Land/Farm Verification → Credit Assessment → Digital/In-kind Disbursement → Loan Monitoring

This transition creates an opportunity for banks to use **alternative data, satellite imagery, digital land records and transaction data** to improve borrower assessment and reach farmers who may have limited access to conventional branch-based banking.

For a **Digital Retail Bank**, agriculture lending can therefore evolve from a traditional collateral- and branch-dependent model toward a **data-driven, digitally delivered financing model**, particularly for small-ticket crop, livestock and input financing.

7. Dashboard Insights

7.1. How large is Pakistan's agriculture financing market?

Agricultural credit disbursement by the 39 banks in our dataset grew from **PKR 1,183.8 billion in 2020 to PKR 2,542.6 billion in 2025**, a compound annual growth rate (CAGR) of **16.5%**. The market more than doubled in five years, with growth every year: 13.3% (2021), 3.7% (2022), 25.4% (2023), 25.3% (2024) and 16.4% (2025).

Habib Bank Limited (HBL) is the market's largest agricultural lender, disbursing **PKR 455.6 billion** in 2025, a **17.9%** market share. Its lending grew every year, from PKR 186.2 billion in 2020, a CAGR of about **19.6%**, with the largest increase in 2025 (+34%).

Leadership has alternated between HBL and the National Bank of Pakistan (NBP): NBP was the largest lender in 2020, 2021 and 2024, and HBL in 2022, 2023 and 2025.

Outstanding agricultural loans (gross advances) grew more slowly, from PKR 669 billion to PKR 1,328 billion (CAGR 14.7%), slightly slower than disbursement, and **agriculture's share of total bank advances slipped from 7.45% to 7.03%**, while total advances grew at 16.0% a year. Agricultural lending is growing at roughly the same pace as overall bank lending.

Note: *Disbursement figures are from the State Bank of Pakistan on a fiscal-year basis (e.g. 2025 = July 2024 to June 2025).*

7.2. Who is currently serving the market?

Commercial banks dominate, providing 78.5% of agricultural disbursement. The market is not highly concentrated: the Herfindahl-Hirschman Index (HHI) is about 870, well below the 1,500 threshold for moderate concentration. Still, the top five lenders account for 56.7% of disbursement. Full Islamic banks are the fastest-growing group (25.0% CAGR), while the microfinance banks' share fell from 11.5% to 9.9% and digital banks hold only 0.05% of the market.

The more revealing pattern is **scale versus focus**. For commercial banks, agriculture is only about 4.4% of their loan books. They are large agri lenders because they are large banks, not because agriculture is a priority.

Microfinance banks are the opposite: agriculture makes up roughly half (52.3%) of their portfolios, which shows that small-ticket agricultural lending can sustain a dedicated business model.

7.3. Where is financing demand concentrated?

Note: *Province-wise lending data is not available, so demand is measured through estimated figures: number of farms, agricultural workforce, cultivated area and crop output.*

Province	Share of farms	Share of Agri workers	Share of cultivated area	Avg farm size
Punjab	41.8%	61.9%	69.7%	6.14 acres
Khyber Pakhtunkhwa	37.1%	16.6%	6.3%	2.04 acres
Sindh	15.7%	16.8%	19.6%	4.80 acres
Balochistan	5.5%	4.6%	4.4%	15.68 acres

- **Punjab is the core of demand.** It leads on almost every indicator and produces 56% of national crop output.

- **Khyber Pakhtunkhwa has a distinct profile.** It holds over a third of the country's farms (4.48 million) on only 6.3% of the cultivated area. These are small farms that need small loans, a segment that is costly to serve through branches.
- **Sindh** has a strong crop base and dominates **fisheries (57% of national output)**, a niche that the Risk Coverage Scheme now explicitly covers.
- **Balochistan** has the highest agricultural share of provincial GDP (~25%), but its farm base is small and its farms are large, which limits the scale of the market.

On average, each farm received about **PKR 210,400** in agricultural credit in 2025, compared with about PKR 114,000 in 2020.

7.4. Where can a Digital Retail Bank participate?

The DRB Opportunity Index scores each province from 0 to 100 using six equally weighted factors: agricultural employees, agriculture's share of GDP, agriculture's share of employment, number of farms, small farm size and cultivated area.

Rank	Province	DRB Opportunity Score (2025)
1	Punjab	85.5
2	Khyber Pakhtunkhwa	67.7
3	Sindh	51.4
4	Balochistan	36.2

Punjab offers the largest market, but it is also the most competitive, with commercial banks, the Bank of Punjab and PPCBL already active. A DRB would need a clear point of difference here, such as speed, a digital journey and input financing through merchants.

Khyber Pakhtunkhwa is the most natural fit for a digital model**.** Its large number of very small farms matches the small-ticket, low-cost lending that digital channels do best. Its score also **rose from 58.9 in 2020 to 67.7 in 2025**, the fastest improvement of any province, as the number of farms grew and average farm size fell. Under the Risk Coverage Scheme, farmers of all landholding sizes in KP are eligible for the 10% first-loss cover, which further reduces lending risk there.

Sindh is a moderate opportunity, with fisheries and crop finance as possible niches.

Balochistan is a longer-term option: Risk Coverage Scheme support is broad, but the market is small.

8. Opportunity for Digital Banks

Microfinance banks have demonstrated that **small-ticket agricultural lending can be commercially viable**. Digital retail banks (DRBs) can build on this model while reducing the cost of serving farmers through **digital onboarding, satellite-enabled verification, alternative data, and merchant-led disbursement**. This

can make agricultural financing more scalable, accessible, and cost-efficient—particularly for small and underserved farmers.

8.1. Highest-Potential Products

#	Agriculture financing product	Reason for a Digital Retail Bank
1	**Digital Crop Production / Input Loan**	Small-ticket, seasonal loans that fit Zarkhez-e's merchant-based disbursement and the Risk Coverage Scheme. It directly serves KP's 4.5 million small farms.
2	**Livestock & Dairy Finance**	Covered by the Risk Coverage Scheme, and daily milk income supports frequent small digital repayments. Bundling with insurance lowers risk.
3	**Electronic Warehouse Receipt Financing (EWRF)**	Low-risk lending because the stored commodity is the collateral and receipts are already electronic. It helps farmers avoid selling at harvest-time low prices.
4	**Poultry Farming Finance**	Short 6–8 week production cycles allow quick repayment and fast credit scoring. Best started with feed and chick financing.
5	**Fisheries & Fish Farming Finance**	A niche in Sindh, which produces 57% of national fisheries output, and covered by the Risk Coverage Scheme.
6	**Irrigation & Solar Pump Finance**	Can be offered as installment plans through equipment vendors, keeping ticket sizes manageable. It improves farm productivity and therefore repayment capacity.

8.2. Strategic Opportunity

The opportunity for a DRB is **not simply to build another loan app for farmers**. The larger opportunity is to create a **digital agricultural finance ecosystem** that supports the farmer throughout the complete financing cycle—from **input purchase and crop production to harvesting, storage, and sale**.

By combining digital onboarding, data-driven credit assessment, remote verification, merchant networks, and ecosystem partnerships, a DRB can potentially serve smaller agricultural borrowers at a **lower cost-to-serve while improving access, monitoring, and repayment efficiency**.

9. Conclusion

Pakistan's agriculture sector is large and economically vital, yet it remains underserved by formal finance, especially among small and underserved farmers. Agricultural credit disbursement has more than doubled since 2020, but agriculture's share of bank balance sheets has slightly declined. For most commercial banks it is still a marginal activity rather than a strategic priority. This leaves a significant gap in the small-ticket segment, which traditional branch-based, collateral-dependent models find costly to serve.

The enabling environment has, however, changed considerably. The Risk Coverage Scheme shares part of the credit risk with the government. Zarkhez-e provides a ready digital channel for applications and merchant-based disbursement, and SBP's acceptance of satellite-based verification reduces the need for manual field checks. The experience of microfinance banks, which hold about half of their portfolios in agriculture, confirms that small-ticket agricultural lending can be commercially viable. A digital model can build on this by serving the same segment at lower cost and greater scale.

For a Digital Retail Bank, the opportunity is therefore both real and timely. Khyber Pakhtunkhwa offers the most suitable entry point because of its:

- large base of very small farms,
- improving opportunity score, and
- full eligibility for first-loss cover.

Punjab remains the largest long-term market for a bank that can differentiate on speed and digital experience. Success will depend on strong merchant networks, data-driven credit assessment and value-chain partnerships. A DRB that moves beyond a standalone loan product toward an integrated agricultural finance ecosystem can capture meaningful market share while advancing financial inclusion and productivity in Pakistan's rural economy.

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